

System Requirements Analysis

1. Short Answer

For LOCAL deployment: 4 CPU / 16GB is NOT enough for comfortable use.

For INFERENCE VIA MODAL: Your local specs do not matter - Modal runs on L4 GPU.

2. Model Memory Requirements (Llama-3-8B + LoRA)

Component	FP16	4-bit QLoRA	Notes
Base Model	16 GB	4.5 GB	Quantized with bitsandbytes
LoRA Adapter	27 MB	27 MB	Negligible
KV Cache	2-8 GB	1-4 GB	Depends on seq length
Activations	4-8 GB	2-4 GB	With gradient checkpointing
Python + OS	2-3 GB	2-3 GB	Fixed overhead
TOTAL	24-35 GB	10-16 GB	

3. Local vs Modal Inference Comparison

Factor	Local (4 CPU / 16GB)	Modal (L4 GPU)
Memory	Tight fit (16GB)	24GB GPU (plenty)
Speed	2-5 min per call	2-5 sec per call
Cost	Free (electricity)	\$0.0003 per call
Availability	Always on	Scales to zero
Setup	Complex	One command

4. Why Local Deployment Is Problematic

With 4 CPU cores and 16GB RAM, the model CAN run in 4-bit quantization but will be extremely slow. Each inference call would take 2-5 minutes on CPU vs 2-5 seconds on an L4 GPU. For a trading bot that needs to make decisions quickly, this is unacceptable.

Additionally, 16GB RAM leaves almost no headroom. The OS and Python need 3-4GB, leaving 12GB for the model. With 4-bit quantization the model needs 10-16GB, so you would likely hit swap space and performance would degrade further.

5. The Designed Architecture

The system is designed to run inference on Modal serverless GPU, NOT locally. Your local machine only runs the trading bot logic (data fetching, risk management, order execution) which uses negligible resources.

Flow:

Local (4 CPU / 16GB) -> Fetches market data from Alpaca

Local -> Sends context to Modal GPU via API

Modal (L4 GPU) -> Runs LLM inference in 2-5 seconds

Modal -> Returns trade decision to local

Local -> Executes trade via Alpaca API

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6. Summary

Your 4 CPU / 16GB machine is perfectly fine for running the trading bot. The LLM inference happens on Modal L4 GPU which you already have set up. Do NOT try to run the model locally - it would be 100x slower and unreliable.